SAFF: A Spatially Aware Fusion Framework for Effective and Efficient Aerial Object Detection

Hongru Xiao, *Graduate Student Member, IEEE*, Jiankun Zhuang, Bin Yang, Jinming Hu, Junze Zhu, Zhen Lian, Sijie Zhao, and Yanmin Zhou, *Member, IEEE*

***Abstract*—Aerial-based object detection aims to accurately locate vehicles, personnel, and other targets, providing critical support for smart cities and traffic management. While some methods have made partial progress in addressing challenging detection scenarios by fusing infrared (IR) and visible (RGB) modalities, challenges remain in optimizing cross-modal fusion efficiency and balancing the computational cost of additional modalities with detection accuracy. These challenges represent significant barriers to the progress of real-time aerial multimodal detection. To address these challenges, this article proposes a lightweight IR-RGB fusion framework designed to enhance cross-modal fusion efficiency and balance the disparity between computational cost and detection accuracy. Specifically, a spatially aware fusion framework (SAFF) based on the characteristics between IR and RGB modalities is designed, enabling efficient processing of cross-modal and cross-scale feature interactions, thereby significantly reducing computational costs. Moreover, a semantic–spatial fusion module (S2FM) is proposed, which leverages spatial location information from the IR modality to enhance the single RGB-based detector, facilitating complementary improvements in both location accuracy and discrimination capabilities. Experimental validation on three multimodal detection datasets [DroneVehicle, VEDAI, and multiscenario multimodality fusion dataset (M3FD)] using Yolov11 and Yolov12 as single-modal baseline detectors shows that the proposed multimodal framework achieves a 12.2%–19.2% reduction in floating-point operations (FLOPs), along with 2.6%–5.3% and 2.7%–7.1% improvements in mAP and mAP_{50}, respectively, demonstrating a superior balance between detection accuracy and computational efficiency.**

***Index Terms*—Aerial object detection, deep learning, infrared (IR)-RGB detection, multimodal fusion.**

I. Introduction

OBJECT detection from an aerial perspective aims to provide accurate target localization over a wide field of view and has become a fundamental technology for smart transportation, urban planning, and disaster-emergency management [1], [2]. Due to the inherent properties of aerial imagery, objects are captured from high altitudes, resulting in small sizes and dense distributions that significantly weaken discriminative appearance cues. Moreover, aerial objects frequently exhibit arbitrary orientations rather than canonical upright poses, imposing higher demands on fine-grained boundary perception and orientation-aware localization [3]. As illustrated in Fig. 1(a), under challenging aerial conditions such as high-altitude capture, low illumination, and small object size, RGB-only detectors often suffer from degraded spatial cues, leading to blurred boundaries and unreliable localization.

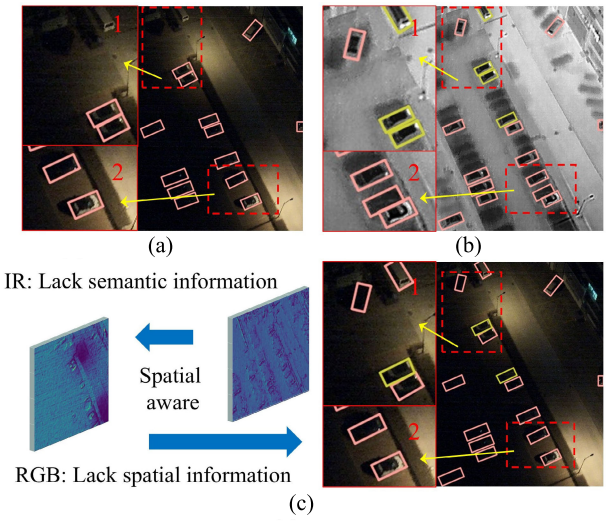


Fig. 1. Purely visible (RGB) image lacks spatial information, making it challenging to accurately localize objects. On the other hand, infrared (IR) modality images lack semantic information, making it difficult to identify objects. By efficiently fusing the semantic and spatial information from both modalities, superior detection results can be achieved. (a) RGB: hard to locate. (b) IR: hard to recognize. (c) RGB: lack spatial information.

IR sensors can detect the intensity of thermal radiation emitted by objects, providing the advantage of robustness against variations in lighting conditions [4]. To this end, exploring the use of IR radiation signals to assist in RGB image detection has emerged as a promising solution [5], [6]. However, due to the absence of background information, such as color, and the presence of thermal crosstalk in thermal sources, the IR modality makes it hard to recognize the objects, as illustrated in Fig. 1(b). Essentially, in aerial imagery detection tasks,

Received 30 January 2026; accepted 9 March 2026. Date of publication 17 March 2026; date of current version 3 April 2026. This work was supported by the National Key Research and Development Program of China under Grant YFD1600400402. *(Hongru Xiao and Jiankun Zhuang contributed equally to this work.) (Corresponding author: Bin Yang.)*

Hongru Xiao, Jiankun Zhuang, Bin Yang, Jinming Hu, and Zhen Lian are with the College of Civil Engineering, Tongji University, Shanghai 200092, China (e-mail: hongru_xiao@tongji.edu.cn; 2510077@tongji.edu.cn; yangbin@tongji.edu.cn).

Junze Zhu is with the School of Computer Science and Technology, Tongji University, Shanghai 200092, China (e-mail: 2351114@tongji.edu.cn).

Sijie Zhao is with the School of Geography and Ocean Science, Nanjing University, Nanjing 210023, China (e-mail: zsj2233@smail.nju.edu.cn).

Yanmin Zhou is with the College of Electronics and Information Engineering, Tongji University, Shanghai 200092, China (e-mail: yanmin.zhou@tongji.edu.cn).

Data is available on-line at https://github.com/Hongru0306/SAFF_det
Digital Object Identifier 10.1109/TGRS.2026.3674467

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:02:06 UTC from IEEE Xplore. Restrictions apply.

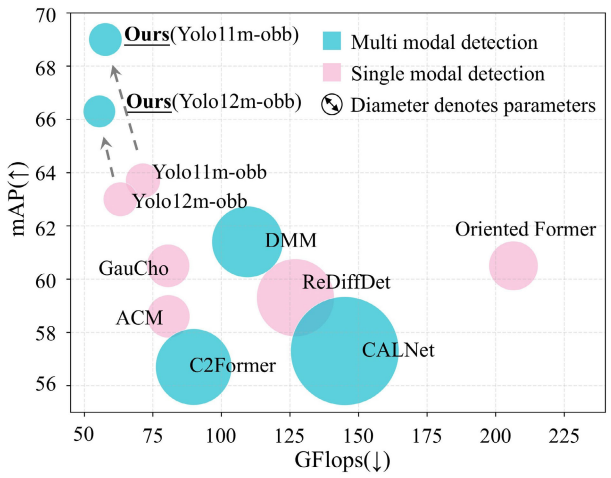


Fig. 2. On the DroneVehicle dataset, our method achieves superior detection performance while incurring lower computational cost compared to the single-modal baseline models. These results demonstrate its effective balance between performance and efficiency in multimodal object detection.

fine-grained object recognition is crucial, requiring detection models to possess both robust localization and recognition capabilities [7]. Therefore, enhancing object detection by combining the rich background semantic information from the RGB modality with the object location information from the IR modality presents an insightful solution [8]. However, with the assistance of the additional IR modality, it has also led to high computational costs, highlighting the inefficiency of current methods and limiting the applicability on resource-constrained aerial platforms [9].

To address these challenges, we propose an SAFF, designed to more efficiently fuse IR and RGB cross-modal information. This approach aims to achieve complementary integration of spatial and semantic information, enhancing detection accuracy while minimizing the computational costs introduced by the additional modality. Specifically, the network framework consists of two core components.

1) During the integration of the IR modality to assist RGB image detection, an SAFF is proposed, which enables the hierarchical fusion of shallow spatial information with deep semantic information, thereby reducing the computational cost introduced by the additional modality.
2) To get better cross-modal fusion performance, we propose an efficient semantic–spatial fusion module (S2FM). It first deeply aggregates the semantic information from the RGB modality and embeds the spatial information from the IR modality. The information from both modalities is then dynamically mapped to a unified representation layer, and finally, the semantic and spatial information are adaptively fused through a learnable adjustment converter, achieving cross-modal information complementarity. As shown in Fig. 1(c), this process ultimately collaborates to enhance the model's detection performance.

Fig. 2 illustrates the main experimental results of SAFF in challenging multimodal aerial scenarios. By leveraging the proposed efficient cross-modal fusion strategy, SAFF achieves a 5.3% improvement in mAP over the baseline single-modal Yolo11m-obb while simultaneously reducing the computational cost by 13.7 GFLOPs. This demonstrates the framework's ability to strike an optimal balance between detection accuracy and efficiency, highlighting its superiority over the SOTA models. Our main contributions are as follows.

1) A spatially aware fusion framework (SAFF) for multimodal aerial object detection is proposed, where cross-modality interaction information is transmitted across different layers and scales, reducing computational costs while enhancing detection accuracy.
2) We propose an efficient cross-modal fusion module, S2FM, which effectively aggregates semantic information and adaptively embeds spatial information to achieve cross-modal semantic–spatial complementarity, thereby further enhancing detection performance.
3) Extensive experiments on the three multimodal datasets demonstrate that SAFF surpasses SOTA detectors while taking a lower computation cost than base single-modal detectors. We also take two different base detectors to validate the plug-and-play nature of SAFF, showing consistent superior performance.

II. Related Work

A. Object Detection in Challenge Scenes

Aerial imagery contains abundant information, but objects in aerial perspectives typically occupy only a small portion of the image or consist of very few pixels [10], [11]. Furthermore, due to the broad field of view, targets in aerial images are often surrounded by complex backgrounds, making their detection challenging. Some researchers have explored ways to enhance the feature extraction process. Yang et al. [12] proposed a joint model integrating keypoint detection and object localization, which enhances the model's capability to determine object association, thereby improving detection performance in complex scenarios. Zhao et al. [13] designed an RSM module that performs multidirectional global modeling for ultrahigh-resolution remote sensing images, enabling more accurate capture of large-scale spatial features. Xiao et al. [14] proposed a feature complementary mapping module that enhances the model's capability for target recognition, thereby improving overall detection performance.

In addition to enhancing the model's feature extraction capability, optimizing the object recognition process also represents an important direction for improving recognition performance. Xie et al. [15] reintegrated decoupled classification and regression features by introducing shared offsets, which effectively reduced the inconsistency between feature predictions. In addition, they jointly employed anchor-based and anchor-free regression schemes, significantly enhancing detection performance across objects with diverse characteristics. Yang et al. [16] performed instance-level denoising on the feature map to improve detection of small and cluttered objects. Xiao et al. [17] introduced a knowledge-injection-based training strategy that guides the model to learn the discriminative boundaries between targets and their surrounding environments, thereby enhancing its capability

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:02:06 UTC from IEEE Xplore. Restrictions apply.

for precise target localization and recognition. Xie et al. [3] proposed DFDet, a dual-focus oriented detector that jointly leverages multidimensional contextual features and mitigates angle regression sensitivity. Key components include a context-dependent mining network for compact, context-rich representations and a penalty-incentive allocation strategy that dynamically adjusts angle regression loss to improve localization of elongated targets.

Nevertheless, RGB-based detection accuracy is affected not only by object size but also by extreme illumination; existing single-modal remote sensing detectors face severe challenges [18], [19], [20]. Additional modalities, such as IR data, due to their imaging physical properties, are less susceptible to interference from lighting and other environmental factors, offering a promising avenue for improving detection accuracy in challenging scenarios [21].

B. IR-RGB Object Detection

IR data capture object location information via thermal signals, offering a complementary sampling perspective to RGB imagery. By integrating IR and RGB modalities, the model can more effectively exploit cross-modal spatial cues, leading to improved object detection performance [9]. Therefore, some researchers have attempted to enrich the original visible-light information by integrating IR modality information into RGB images. Zhang et al. [22] proposed a novel adversarially trained differential feature-aware network that fuses attention-guided features from IR and visible modalities, effectively enhancing detection performance. Lv et al. [23] designed an edge extraction network for low-light visible images, which can accurately capture critical structural information and inject it into the fused image, thereby improving cross-modal image matching performance under challenging conditions. Zhao et al. [24] combined a hybrid transformer with a convolutional neural network encoder architecture and proposed a novel composite attention fusion strategy, which effectively integrates global and local information to enhance multimodal image fusion performance.

By performing complementary fusion enhancement across modalities, the information from both modalities can be mutually reinforced. However, for specific detection tasks, dedicated task-oriented optimization and design are still required. Yuan and Wei [21] designed an ICA module that learns the cross-modal attention relationships between RGB and IR modalities to obtain calibrated and complementary features, while an adaptive feature sampling module is further introduced to reduce the computational complexity of feature interaction. Zhou et al. [25] proposed a disparity-guided multispectral Mamba model that adaptively fuses RGB and IR image features by leveraging disparity information between different modalities, effectively mitigating cross-modal conflicts and enhancing detection performance. Li et al. [26] proposed a bidirectionally driven saliency fusion network that enhances the contrast between the target and background while fully exploiting the complementary characteristics of source images in both spatial and frequency domains, thereby jointly improving the model's capability for multimodal object detection.

Although multimodal detection optimization has achieved improvements in detection performance, there remains a considerable gap in balancing detection accuracy and computational cost compared with single-modality detection models [14]. Therefore, the goal of our study is to enhance detection accuracy while significantly reducing computational cost, potentially achieving efficiency even lower than that of single-modal baseline detection models.

C. Research Gaps

In aerial multimodal detection, object instances are typically small and densely distributed, with severely weakened visual cues, which makes accurate object localization and fine-grained boundary perception particularly critical. In this context, multimodal information, particularly IR signals, provides not only complementary features but also valuable spatial and boundary cues. More importantly, aerial multimodal detection places greater emphasis on efficiently exploiting such spatial cues from the IR modality to guide and enhance semantic feature learning in the RGB modality, thereby improving detection robustness and localization accuracy under severe appearance degradation.

Additionally, many generic multimodal detection methods incorporate IR as an auxiliary modality to improve localization robustness, but this often incurs a significant increase in computational cost due to extra feature extraction and cross-modal fusion. Such heavy fusion designs generally overlook the strict efficiency constraints and spatial redundancy inherent in aerial imagery with small and densely distributed targets [9], [24].

Motivated by these observations, this work focuses on designing an efficient multimodal detection framework tailored for aerial imagery. The proposed approach selectively leverages spatial and boundary cues from the IR modality to guide semantic feature learning in the RGB modality while explicitly suppressing redundant cross-modal interactions. By aligning fusion design with the efficiency constraints and spatial characteristics of aerial scenes, our goal is to achieve high detection accuracy with minimal additional computational overhead.

III. Method

A. Spatially Aware Fusion Framework

As illustrated in Fig. 3, the proposed SAFF jointly exploits RGB and IR modalities by embedding multiscale and cross-depth multimodal interaction modules into both the backbone and the neck architectures.

1) Cross-Modal Fusion Formulation: In the proposed fusion modules, features rich in spatial information from the IR modality I_i are integrated with features carrying semantic information from the RGB modality I_r. This design enables the network to enhance RGB semantic representations with complementary IR spatial cues. The fusion process can be formally expressed as

$$F_f = \Phi(I_r, I_i; P_c) \tag{1}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:02:06 UTC from IEEE Xplore. Restrictions apply.

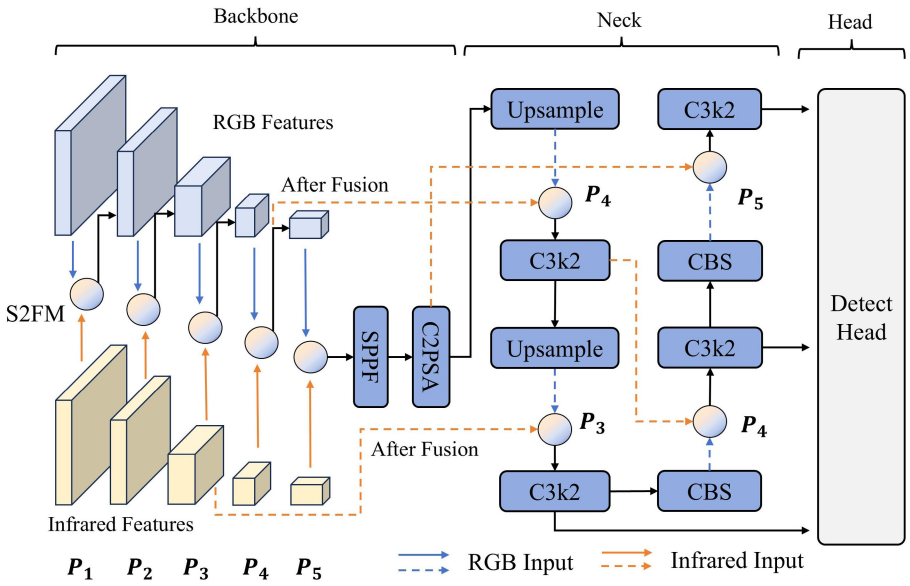


Fig. 3. Overall of the SAFF (Yolo11). The circles represent the S2FMs. For different detectors such as Yolo12, the overall framework is similar, just need to replace the corresponding updated modules such as SPPF and C3k2.

where F_f denotes the fused feature map, $\Phi(\cdot)$ represents the proposed cross-modal fusion module S2FM, and P_c indicates the stage at which spatial resolution changes occur due to downsampling or upsampling operations.

2) Backbone Feature Fusion: The backbone network consists of five hierarchical downsampling stages, denoted as P_1–P_5. At each stage P_k, the RGB and IR feature maps at the same spatial resolution, $\{I_r^k, I_i^k\}$, are jointly processed by the S2FM module. The resulting fused feature F_f^k serves as the RGB feature for the subsequent stage, being downsampled and forwarded through the RGB branch to the next S2FM module at P_{k+1}. Simultaneously, the IR feature I_i^k is downsampled along the IR branch and fused with F_f^k at the next stage. In this way, cross-modality interaction information is progressively accumulated across layers and scales, enabling the backbone to effectively capture both low-level spatial details and high-level semantic features.

3) Neck Cross-Scale Fusion: In the neck, dashed blue and orange lines indicate the RGB and IR input branches, respectively. S2FM modules perform cross-depth fusion at three spatial resolutions, corresponding to stages P_3–P_5. At each stage, upsampled deep features from higher levels are fused with the corresponding shallow fused features output by the backbone S2FM modules. Shallow features, containing fine-grained spatial and positional information, are combined with deeper semantic features at matching resolutions to produce spatially aware representations. This design enables the model to capture complementary spatial and semantic information across depths, enhancing its utilization of multimodal features for improved detection performance.

4) Channel Width Reduction: To further optimize the architecture of the multimodal detection model, the proposed model reduces the channel width for both the RGB and IR branches in the backbone, complemented by a multilevel fusion architecture. This approach not only enhances cross-modal interaction and perception but also reduces the computational cost of the model. The computation process is as follows:

$$C_{\text{num}} = \mathcal{H}(M_n), \quad P_1 \le n \le P_5 \tag{2}$$

where $\mathcal{H}$ represents the operation that halves the channel count and M_n denotes the number of channels at the nth module in the backbone. The channel reduction operation is specifically applied to the modules between $P_1 \le n \le P_5$ in the backbone network.

By incorporating cross-scale spatial information awareness interaction layers and reducing redundant feature extraction modules in the network architecture, SAFF emphasizes the complementary interactions between modalities. This design enables more effective multimodal feature fusion, facilitating information complementarity while simultaneously reducing the computational cost of the model.

B. Semantic–Spatial Fusion Module

The S2FM module aims to fully integrate the positional information from the IR modality with the semantic information from the visible light modality. By combining and complementing positional and semantic information, the module enhances the model's ability to accurately locate and recognize objects in challenging scenarios. As shown in Fig. 4, S2FM consists of four main components: the aggregation of semantic information from the visible light modality, the embedding of spatial information from the IR modality, the cross-modal fusion block, and the final modality interaction fusion.

1) Semantic Information Aggregation: In this step, the deep aggregation module is employed to extract semantic information from the RGB modality. Given an input RGB image $I_r \in \mathbb{R}^{C\times w\times h}$, a local adaptive average pooling operation is applied to project the semantic feature maps to a unified spatial resolution, following commonly adopted pooling strategies for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:02:06 UTC from IEEE Xplore. Restrictions apply.

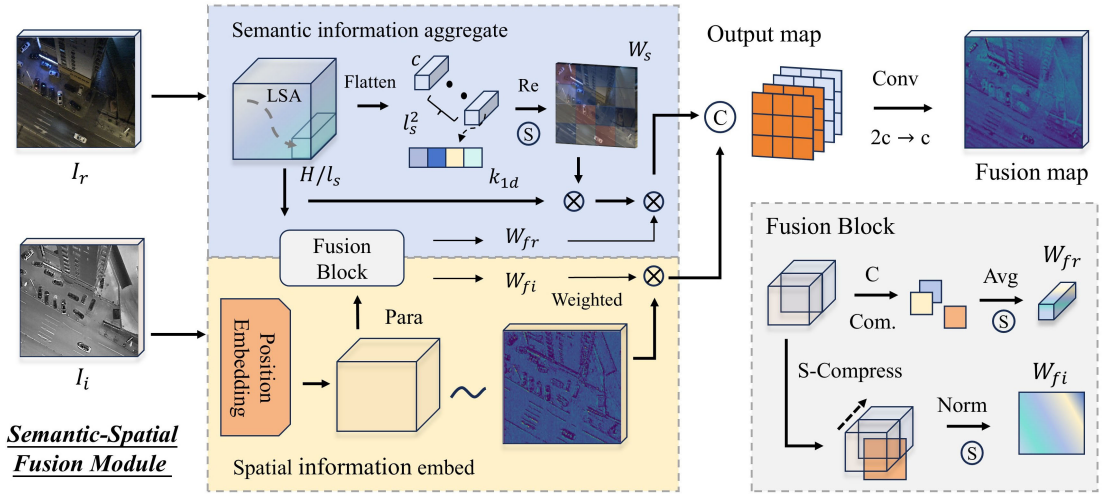


Fig. 4. S2FM architecture. RGB features I_r and IR features I_i are separately processed by the semantic information aggregate and spatial information embed modules, respectively, and then dynamically fused by the fusion block to enable cross-modal interaction.

efficient feature aggregation [27]. The resulting feature map S is expressed as

$$S = \mathcal{A}_{l_s}(I_r) \in \mathbb{R}^{C \times l_s \times l_s} \tag{3}$$

where $\mathcal{A}_{l_s}$ denotes the local aggregation operator, which maps the spatial dimensions $w \times h$ of I_r to $l_s \times l_s$ by averaging over adaptive local regions, producing a compact and semantically rich representation for subsequent local-domain attention modeling.

Next, the perceptual features obtained from local aggregation are further processed to model cross-channel interactions among neighboring channels. Specifically, the feature map $S \in \mathbb{R}^{C \times l_s \times l_s}$ is reshaped into a 1-D sequence while preserving the channel ordering. A 1-D convolution is then applied along this sequence so that each output response is computed from a local neighborhood of adjacent channels.

The corresponding computation is formulated as

$$a = \text{Conv1d}\left(\text{Flatten}_{1 \times (C \cdot l_s^2)}(S), k_{1d}\right) \tag{4}$$

where k_{1d} is a learnable parameter in the 1-D convolution, dynamically determined based on the input feature maps.

After enhancing local correlations, the aggregated features are normalized and reconstructed to obtain $A \in \mathbb{R}^{C \times l_s \times l_s}$, which encodes region-wise semantic responses over compact local spatial regions.

To align these local semantic responses with the spatial resolution of the target feature map, adaptive average pooling is employed to resize A to the RGB feature size $C \times w \times h$. Accordingly, the semantic aggregation weight at each spatial location (i, j) is computed as the average response within its corresponding local region $R_{i,j}$

$$W_s(c, i, j) = \frac{1}{|R_{i,j}|} \sum_{(u,v) \in R_{i,j}} A(c, u, v) \tag{5}$$

where $R_{i,j}$ denotes the input region adaptively determined by the pooling operation.

Finally, the refined semantic feature F_r is obtained by applying the resulting attention map to the original input features via element-wise multiplication

$$F_r = W_s \odot I_r. \tag{6}$$

This operation refines the semantic representation by emphasizing the most relevant spatial and channel-wise features for subsequent tasks, ultimately enhancing performance in multimodal object detection and recognition.

2) Spatial Information Embed: In order to more effectively capture the spatial information of objects in the IR modality I_i, a point-wise convolution operation is applied to extract a feature map that encodes the spatial object information within the image. This approach efficiently preserves the shallow spatial positional information of the objects while further stabilizing the spatial feature representation via normalization and activation functions. This process is represented by the following equation:

$$S_e = \phi\left(I_i, k_{1\times1}\right) \tag{7}$$

where ϕ represents the learnable spatial information embedding operator. The output feature map S_e retains the same shape as the input, $\mathbb{R}^{C \times W \times H}$.

3) Cross-Modality Fusion: To fully enhance the interaction between modalities, we first concatenate the raw features from both modalities to obtain the preliminary fused information $W_f \in \mathbb{R}^{2C \times W \times H}$. Subsequently, cross-modal complementary mappings are performed on the semantic feature F_i and the spatial feature F_r, facilitating the initial interaction between semantic and spatial information. Specifically, starting with the spatially embedded feature map F_r, the raw cross-modality information W_f is integrated across the channel dimension to generate adjustment weights for the spatial information. The computation is given by the following equation:

$$W_f^{'(1,x,y)} = \sum_{c=1}^{2C} W_k(1, c, 1, 1) \cdot W_f(c, x, y) \tag{8}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:02:06 UTC from IEEE Xplore. Restrictions apply.

where W_k represents the learnable weights, and the resulting W'_f is processed through normalization and sigmoid function to generate the final spatial attention weights $W_{fi} \in \mathbb{R}^{1\times W\times H}$.

Similarly, by applying spatial mapping across the spatial dimension of the concatenated raw feature W_f, we align the channel fusion information with the semantic feature F_r that has been aggregated. This alignment is further refined through pooling and activation functions, providing the semantic-level fusion attention weights $W_{fi} \in \mathbb{R}^{C\times 1\times 1}$. The computation is expressed as follows:

$$W_{fi} = \varphi\left(W_f, k^{2C\to C}\right) \tag{9}$$

where φ represents the semantic weight mapping converter and $k^{2C\to C}$ denotes the learnable parameter that aligns the channel-level fusion information.

4) Fused Feature Output: After obtaining the corresponding feature and coordination weights, these weights are mapped onto their respective features. The two are then connected together, and the final information fusion is performed along the channel dimension to achieve cross-modal spatial–semantic information complementarity. The computation is given by the following equation:

$$F_f = \psi\left(\mathrm{Cat}\left(W_{fr}\odot F_r, W_{fi}\odot F_i\right), k^{2C\to C}\right) \tag{10}$$

where ψ is a learnable channel-wise information mapping converter and k represents the learnable parameters. The final fused feature map $F_f \in \mathbb{R}^{C\times W\times H}$.

Overall, by facilitating cross-modal information interaction between visible and IR images, semantic and spatial information complement each other, thereby enhancing the model's object detection capabilities in challenging detection scenarios. Furthermore, in SAFF, the S2FMs are distributed across multiple scales within the detector. At deeper scales, shallow fusion information progressively strengthens the spatial–semantic interaction and aggregation process as the feature layers deepen. This shallow aggregated information is then used as a spatial embedding branch, continuously supplementing the model's spatial awareness. For the implementation pseudocode of the S2FM, please refer to Algorithm 1.

Algorithm 1 Semantic–Spatial Fusion Module

Input: Visible image feature map I_r, IR image feature map I_i
Begin:

1. **Semantic Information Aggregation**
   $S = \mathcal{A}_{l_s}(I_r) \in \mathbb{R}^{C\times l_s\times l_s}$ ← Semantic feature aggregate;
   $a = \mathrm{Conv1d}(\mathrm{Flatten}_{1\times(C\cdot l_s^2)}(S); k_{1d})$ ← Cross-Channel interaction modeling;
   $A = \sigma(\mathrm{Reshape}(a)) \in \mathbb{R}^{C\times l_s\times l_s}$ ← Get the semantic attention map;
   $W_s(c,i,j) = \frac{1}{|R_{i,j}|}\sum_{(u,v)\in R_{i,j}} A(c,u,v)$ ← Adaptive pooling the attention weights;
   $F_r = W_s \odot I_r$ ← Get the semantic feature;
2. **Spatial Information Embed**
   $S_e = \phi(I_i, k_{1\times 1})$ ← Encode the position information;
   $F_i = \mathrm{Act}(\mathrm{BN}(S_e))$ ← Get the spatial feature;
3. **Cross Modality Fusion**
   $W_f = \mathrm{Cat}(F_r, F_i)$ ← Get the initial fusion weights;
   $W_f^{'(1,x,y)} = \sum_{c=1}^{2C} W_k(1,c,1,1)\cdot W_f(c,x,y)$ ← Spatial-level compress;
   $W_{fi} = \sigma(\mathrm{BN}(W'_f))$ ← Get the spatial information weights;
   $W_{fr} = \varphi(W_f, k^{2C\to C})$ ← Get the semantic information weights;
4. **Fused Feature Output**
   $F_f = \psi(\mathrm{Cat}(W_{fr}\odot F_r, W_{fi}\odot F_i), k^{2C\to C})$ ← Information fusion;

   Return: The final output F_f

End

IV. Experiments and Results

A. Datasets

We conducted experiments on three widely used multimodal object detection datasets: 1) DroneVehicle [18]; 2) VEDAI [37] for oriented bounding box (obb) detection; and 3) multiscenario multimodality fusion dataset (M3FD) [38] for cross-task horizontal bounding box (HBB) detection.

1) DroneVehicle: DroneVehicle is a large-scale drone-based RGB-IR cross-modal benchmark designed for vehicle detection under challenging illumination conditions. It addresses the performance degradation of RGB-only detectors in low-light and nighttime scenarios. The dataset covers diverse real-world scenes, including urban roads, residential areas, parking lots, rural roads, and highways, and spans the full diurnal cycle (day, night, and dark night). In total, 953 087 objects from five categories (car, truck, bus, van, and freight car) are annotated with OBBs, with approximately 49.7% of the samples collected at night or under dark-night conditions.

2) VEDAI: VEDAI is a public benchmark for detecting extremely small vehicles in unconstrained aerial imagery. The dataset contains over 3700 annotated instances across nine fine-grained categories, including car, truck, pick-up, van, boat, plane, camping car, tractor, and others. On average, targets occupy only 0.7% of the image area, posing a significant challenge well beyond the typical object scales in COCO or PASCAL VOC.

3) Multiscenario Multimodality Fusion Dataset: M3FD is a large-scale, well-synchronized RGB-IR dataset for multimodality object detection. It comprises 4177 precisely co-registered image pairs captured by binocular visible and long-wave IR sensors. The dataset spans four scenarios: daytime, overcast, nighttime, and challenging conditions. In total, 33 603 object instances from six common surveillance and autonomous driving categories are annotated with axis-aligned bounding boxes.

B. Compared Models

To evaluate the effectiveness of the proposed SAFF, we conduct comprehensive comparative experiments against a range of representative baseline methods on both IR-RGB OBB and HBB detection tasks.

For the OBB detection task, the compared methods include Yolo11-obb [34], Yolo12-obb [35], OrientedRCNN [28], OAN-OrientedRCNN [29], ReDiffDet [30], ACM-FCOS [31], OrientedFormer [32], GauCho-FCOS [33], UA-CMDet [18], CALNet [36], C2Former [21], and DMM [25].

For the HBB detection task, the compared methods include Yolo11 [34], Yolo12 [35], RT-DETR [39], RT-DETRv2 [40],

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:02:06 UTC from IEEE Xplore. Restrictions apply.

TABLE I
MAIN EXPERIMENT ON DRONEVEHICLE DATASET

Detectors	Params	FLOPs	Car (%)	Truck (%)	Bus (%)	Van (%)	Freight car (%)	mAP (%)	mAP_{50} (%)	Modal
OrientedRCNN [28]	41.4M	91.1G	72.2	57.5	71.4	45.3	42.4	57.8	72.4	
OAN-OrientedRCNN [29]	46.0M	91.1G	72.7	56.8	71.8	46.1	42.9	58.1	72.8	
ReDiffDet [30]	104.6M	127.0G	74.0	54.1	72.1	50.4	46.1	59.3	74.2	
ACM-FCOS [31]	31.9M	80.7G	79.5	54.8	72.9	44.9	40.9	58.6	74.8	
Oriented Former [32]	42.1M	206.5G	76.5	54.6	73.5	51.4	46.3	60.5	75.4	RGB
GauCho-FCOS [33]	31.9M	80.6G	79.7	54.7	72.8	50.5	44.5	60.5	75.9	
Yolo11n-obb [34]	2.7M	**6.6G**	78.3	55.0	73.5	43.0	41.5	58.3	75.5	
Yolo11m-obb [34]	20.9M	71.4G	80.3	63.1	75.3	50.6	48.1	63.7	79.9	
Yolo12m-obb [35]	20.4M	63.2G	80.4	62.3	73.2	51.3	47.4	63.0	78.1	
OrientedRCNN [28]	41.4M	91.1G	71.0	54.4	69.8	44.3	40.0	55.9	70.4	
OAN-OrientedRCNN [29]	46.0M	91.1G	71.6	54.9	70.1	44.9	40.6	56.4	71.0	
ReDiffDet [30]	104.6M	127.0G	72.1	52.2	71.1	42.8	42.7	56.2	71.1	
ACM-FCOS [31]	31.9M	80.7G	77.2	50.8	71.2	43.1	44.9	57.5	72.5	
Oriented Former [32]	42.1M	206.5G	75.9	49.7	74.2	45.7	41.7	57.4	74.0	IR
GauCho-FCOS [33]	31.9M	80.6G	78.0	48.0	73.7	46.6	45.2	58.3	74.3	
Yolo11n-obb [34]	2.7M	6.6G	69.8	51.4	72.3	44.4	41.1	55.8	74.8	
Yolo11m-obb [34]	20.9M	71.4G	72.4	58.4	70.8	51.6	49.7	60.7	79.4	
Yolo12m-obb [35]	20.4M	63.2G	73.8	57.5	71.9	49.5	47.4	60.0	77.3	
UA-CMDet [18]	138.7M	118.2G	75.3	51.9	72.0	41.2	43.3	56.8	71.2	
CALNet [36]	204.0M	145.0G	76.4	52.7	72.5	42.9	43.1	57.3	75.2	
C2Former [21]	100.8M	89.9G	74.8	53.2	70.1	43.2	42.2	56.7	73.3	
DMM [25]	88.0M	109.5G	79.6	60.1	73.3	47.0	47.1	61.4	77.1	IR+RGB
Ours(Yolo11n-obb)	**2.2M**	6.8G	82.3	61.6	76.5	50.8	47.2	63.5	79.2	
Ours(Yolo11m-obb)	19.1M	57.7G	**84.1**	**67.3**	**80.0**	**58.2**	**55.2**	**69.0**	**83.3**	
Ours(Yolo12m-obb)	18.0M	55.5G	83.5	65.5	79.2	52.5	50.6	66.3	80.8	

DEIM [41], D-FINE [42], GM-DETR [43], ICA-Fusion [44], and EI2Det [45].

All compared models are trained and evaluated using the default settings from their official implementations, ensuring fairness and consistency across all experiments.

C. Implementation Details

All experiments were conducted on a workstation equipped with an NVIDIA GeForce RTX 3090Ti GPU. The deep learning models were developed using the PyTorch framework (v2.3.0) with CUDA 11.8 acceleration. Training was performed for 400 epochs with a batch size of 16. Model optimization was carried out using the stochastic gradient descent algorithm, incorporating a momentum term of 0.9 and a weight decay parameter of 0.0001.

For the DroneVehicle dataset, we adopted a label-correction training strategy following the approaches described in [25] and [46]. Specifically, labels generated under the RGB view were used to evaluate the auxiliary effect of incorporating the IR modality on the original RGB-based detection, while conversely, labels derived from the IR view were employed to assess the alignment assistance provided by the RGB modality.

The evaluation metrics include the dataset default settings, mAP, mAP_{50}, Params, and floating-point operations (FLOPs). Following the COCO evaluation protocol, mAP50 refers to the mAP computed at an IoU threshold of 0.5. To further emphasize localization quality, we adopt mAP50:95, which averages AP values over ten IoU thresholds ranging from 0.50 to 0.95 with a step size of 0.05. In this work, mAP50:95 is reported as mAP to provide a concise and widely accepted evaluation of detection accuracy

$$\text{mAP} = \frac{1}{10}\sum_{t=1}^{10} \text{AP}\left(\text{IoU} = 0.50 + 0.05\,(t-1)\right). \tag{11}$$

In addition to accuracy metrics, we report the number of model parameters (Params) and the computational complexity measured in FLOPs to evaluate model efficiency.

Params denote the total number of learnable parameters in the model and can be expressed as

$$\text{Params} = \sum_{i=1}^{N} |\theta_i| \tag{12}$$

where θ_i represents the parameter tensor of the ith layer and N is the total number of learnable layers.

FLOPs measure the total number of FLOPs required for a single forward pass, serving as an indicator of computational cost. The overall FLOPs can be written as

$$\text{FLOPs} = \sum_{l=1}^{L} \text{FLOP}_l \tag{13}$$

where FLOP_l denotes the number of FLOPs in the lth layer and L is the total number of layers.

D. Results on DroneVehicle Dataset

1) Quantitative Results: Our method is compared with current SOTA models for both multimodal and single-modal OBB detection tasks. As shown in Table I, when evaluating the improvement brought by the introduction of the IR modality to the RGB single-modal detection task, SAFF achieves the best detection performance with the more lightweight.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:02:06 UTC from IEEE Xplore. Restrictions apply.

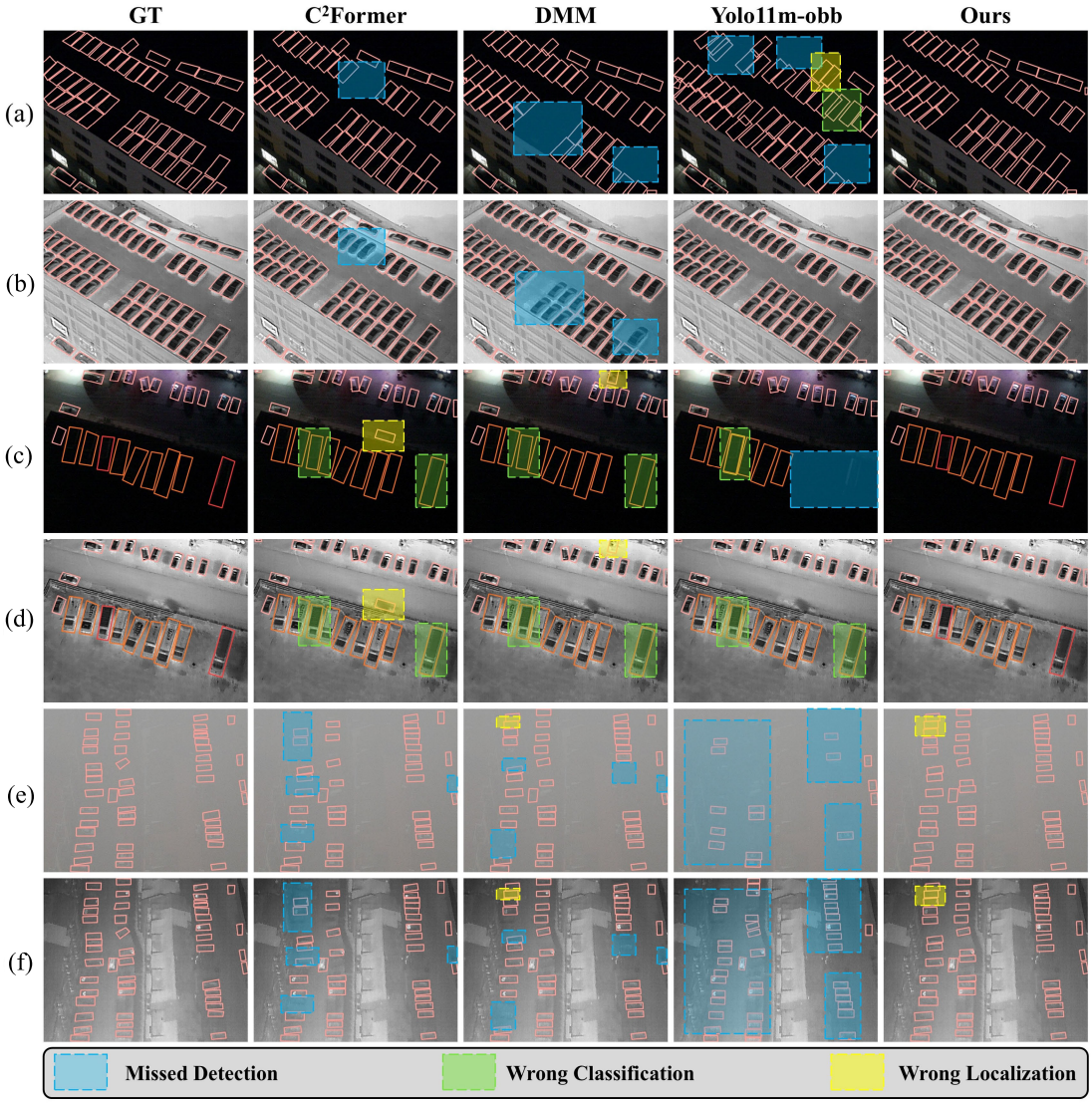


Fig. 5. Visualizationof detection results on the DroneVehicle dataset. The model used for visualization is SAFF (Yolov11m-obb).

Notably, with the proposed optimization strategy for the fusion network architecture, SAFF, even with the addition of the IR modality, exhibits 12.2%–19.2% lower computation cost (FLOPs) than the baseline single-modal detection models, Yolov11m-obb and Yolov12m-obb. Moreover, it leads to a 3.3%–5.3% improvement in mAP, also surpassing other single-modal comparison algorithms, thus highlighting the balance between performance and efficiency in our approach. Besides, in comparison to the compared multimodal detection models, SAFF achieves an improvement of 6.6%–12.2% in mAP with less than 25% of the model Params, demonstrating the efficiency and effectiveness of the proposed method.

2) Visualization Results: To visually demonstrate the model's performance, we conducted a study to visualize the detection results. As shown in Fig. 5, it illustrates that the proposed SAFF achieves the best localization and classification. Specifically, the following conditions hold.

In example [Fig. 5(a) and (b)], our proposed method demonstrates superior robustness in detecting vehicles within shadows, while the comparison models struggled to varying degrees under such conditions. Meanwhile, the comparative models C2Former and DMM produced a higher number of missed detections in low-light conditions. And the single-modal baseline model Yolo11m-obb suffers from missed detection a lot. It also exhibits wrong classification and wrong localization because the RGB modality lacks sufficient positional information.

Example [Fig. 5(c) and (d)] reveals that all competing models, Yolo11m-obb, DMM, and C2Former, produce incorrect category predictions. The IR modality lacks semantic information, leading to the wrong classification of the single-modal baseline model Yolo11m-obb. Notably, the RGB-based Yolo11m-obb model overlooks multiple cars present in regions with heavy shadows. In contrast, the proposed SAFF highlights its superior performance in low-light conditions.

In example [Fig. 5(e) and (f)], the proposed SAFF exhibits a single instance of wrong localization in a challenging case, which also challenges the multimodal method DMM. Additionally, the DMM suffers from multiple missed detections. As for the other two comparative models, they exhibit numerous

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:02:06 UTC from IEEE Xplore. Restrictions apply.

TABLE II

RESULTS ON THE VEDAI DATASET. "$\text{MODAL}_{\text{BEST}}$" DENOTES THE MODALITY WHERE THE SINGLE-MODAL DETECTOR ATTAINS ITS BEST PERFORMANCE

Detectors	FLOPs	mAP (%)	mAP_{50} (%)	$\text{Modal}_{\text{best}}$
ReDiffDet [30]	127.0G	38.9	57.0	RGB
ACM-FCOS [31]	80.7G	37.6	55.5	
Oriented Former [32]	206.5G	38.0	56.2	
GauCho-FCOS [33]	80.6G	37.9	56.1	
Yolo11n-obb [34]	**6.6G**	31.5	48.2	
Yolo11m-obb [34]	71.4G	40.5	58.7	
Yolo12m-obb [35]	63.2G	38.8	55.5	
CALNet [36]	145.0G	35.8	55.7	IR+ RGB
C2Former [21]	89.9G	35.9	56.2	
DMM [25]	109.5G	37.9	57.2	
Ours(11n-obb)	6.8G	36.4	55.3	
Ours(11m-obb)	57.7G	**43.3**	**63.3**	
Ours(12m-obb)	54.4G	42.1	60.8	

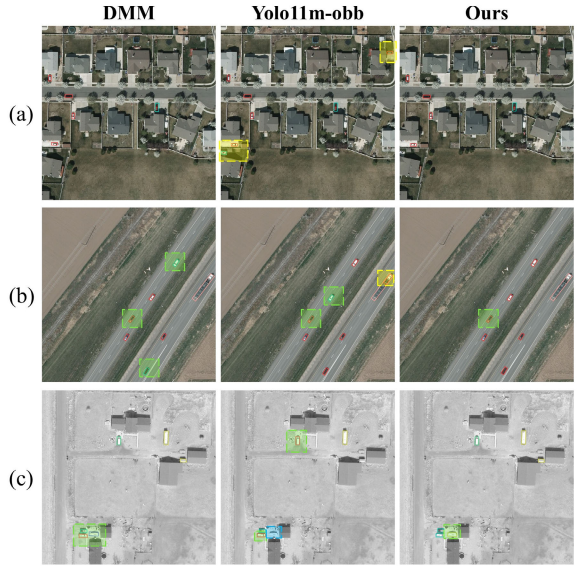


Fig. 6. Visualization of detection results on the VEDAI dataset.

missed detections in multitarget scenarios. The Yolo11m-obb model, in particular, is severely affected by this issue.

E. Results on VEDAI Dataset

To demonstrate the generalizability of the proposed method, we also conducted tests on the VEDAI dataset. As shown in Table II, SAFF achieves a 4.6%–7.1% and 2.8%–4.9% improvement in mAP_{50} and mAP over the baseline, respectively, while also outperforming other state-of-the-art fusion algorithms in terms of both detection accuracy and computational efficiency.

The visual analysis of the detection results of the VEDAI dataset is presented in Fig. 6, the single-modal baseline model Yolo11m-obb, and the multimodal method DMM are compared with our proposed SAFF. In example [Fig. 6(a)], the open field of view posed no challenge to the two multimodal methods. However, Yolo11m-obb produced instances of erroneous localization. In example [Fig. 6(b)], all models, including our proposed SAFF, were susceptible to the missed classification of small objects. The proposed method, however, demonstrated superior robustness in these scenarios, with the DMM being the most severely affected, followed by Yolo11m-obb, which also exhibited significant classification errors. In example [Fig. 6(c)], the proposed SAFF exhibits a single instance of wrong classification in a challenging case, which also challenges the multimodal method DMM. The DMM exhibits more wrong classification than the proposed SAFF. Meanwhile, the single-modal baseline model Yolo11m-obb exhibits both missed detection and wrong classification, further proving the superiority of our model in the context of aerial object detection.

TABLE III

EXPERIMENT ON THE CROSS-TASK (HBB) M3FD DATASET

Detectors	FLOPs	mAP (%)	mAP_{50} (%)	$\text{Modal}_{\text{best}}$
RTDETR [39]	60.3G	51.1	79.2	RGB
RTDETRv2 [40]	92.5G	53.3	82.7	
DEIM [41]	24.8G	51.2	80.2	
D-FINE [42]	24.8G	50.5	79.6	
Yolo11n [34]	6.3G	50.9	76.7	
Yolo11m [34]	68.0G	57.3	83.2	
Yolo12m [35]	59.8G	57.1	83.7	
GM-DETR [43]	176.0G	52.5	82.6	IR+ RGB
ICA-Fusion [44]	108.2G	53.7	85.0	
EI2Det [45]	120.0G	55.0	83.9	
Ours (Yolo11n)	6.5G	55.3	83.0	
Ours (Yolo11m)	55.6G	**60.1**	**87.4**	
Ours (Yolo12m)	53.4G	59.7	86.8	

F. Results on M3FD Dataset

To further assess the generalization capability of SAFF, experiments were conducted on the cross-task HBB multimodal detection dataset M3FD. The comparative baselines consisted of the HBB implementations of Yolov11 and Yolov12. The results in Table III demonstrate that SAFF achieves a 3.1%–6.3% and 2.6%–4.4% improvement in mAP_{50} and mAP, respectively. Notably, when high-performance detectors such as Yolo11m and Yolo12m are adopted as the base models, SAFF incurs even lower computational cost (FLOPs) than their corresponding single-modal counterparts. Similarly, compared to the traditional multimodal methods, SAFF demonstrates a 4.7%–7.6% improvement in mAP with less than 50% of the computational cost (FLOPs), further highlighting the superiority of SAFF in both detection performance and efficiency.

G. Ablation Study

To fully assess the effectiveness of the proposed method, we conduct a series of ablation experiments using the DroneVehicle dataset. Yolov11m-obb is used as the baseline model in all the ablation experiments.

1) RGB Assist IR Detection: To further examine the complementary relationship between the IR and RGB modalities leveraged by the proposed framework, we carried out additional experiments on the DroneVehicle dataset, where IR-based annotations were employed to analyze the contribution of RGB information to IR performance. The results in Table IV indicate that SAFF yields a 2.5%–2.9% improvement in mAP accompanied by a consistent decrease in computational cost, thereby reinforcing the validity of the proposed method.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:02:06 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ABLATION STUDY ON IR-BASED LABELS IN DRONEVEHICLE

Detectors	FLOPs	mAP(%)	mAP_{50} (%)	$Modal_{best}$
ReDiffDet [30]	127.0G	59.9	75.6	IR
ACM-FCOS [31]	80.7G	59.6	75.5	
Oriented Former [32]	206.5G	58.9	74.2	
GauCho-FCOS [33]	80.6G	61.3	76.1	
Yolo11m-obb [34]	71.4G	67.0	81.6	
Yolo12m-obb [35]	63.2G	66.2	81.1	
CALNet [36]	145.0G	59.4	74.7	IR+ RGB
C2Former [21]	89.9G	59.3	75.0	
DMM [25]	109.5G	64.4	79.4	
Ours (11m-obb)	57.7G	**69.9**	**84.2**	
Ours (12m-obb)	**55.5G**	68.7	83.2	

TABLE V
ABLATION STUDY ON KEY COMPONENTS

Fra.	Sem.	Spa.	FB.	Params	FLOPs	mAP (%)	mAP_{50} (%)
-	-	-	-	20.9M	71.4G	63.7	79.9
✓	-	-	-	**18.2M**	**52.2G**	64.5	80.3
✓	✓	-	-	18.3M	52.5G	65.3	80.9
✓	-	✓	-	19.0M	56.9G	65.5	81.2
✓	-	✓	✓	19.1M	57.4G	66.5	81.6
✓	✓	-	✓	18.3M	53.0G	65.9	81.3
✓	✓	✓	-	19.0M	57.2G	67.4	82.3
✓	✓	✓	✓	19.1M	57.7G	**69.0**	**83.3**
✓	IR	RGB	-	18.9M	56.8G	64.5	80.6
✓	IR	RGB	✓	19.1M	57.4G	65.8	81.0

2) Ket Component Effect: To thoroughly validate the impact of each component of the proposed method on the overall performance, we conducted a series of ablation experiments on the core improvement modules. As shown in Table V, the "Fra." denotes the original multimodal framework enhancement without the inclusion of the S2FM modules, and the S2FM modules consist of three key components: "Sem." refers to the semantic information aggregation, "Spa." corresponds to the spatial information embedding, and "FB." represents the fusion block. The results show that the baseline model architecture was first optimized to support dual-modal input, resulting in a reduction of the FLOPs by 26.8% and an improvement of 0.8% in mAP. Additionally, the introduction of the cross-modality fusion block led to a 1.6% improvement in mAP detection performance. Further experiments were conducted on the semantic and spatial branches within the S2FM module. By incorporating the semantic information aggregate enhancement strategy, the network achieved a 2.5% increase in mAP. Furthermore, the spatial information embedding for the IR modality resulted in an additional 3.1% improvement in detection performance.

Besides, we explored the scenario where the IR and RGB data are input to the semantic and spatial branches, respectively, which is the reverse of the original S2FM configuration. The comparative experiment reveals that feeding the IR modality into the spatial embedding module and the RGB modality into the semantic aggregation module (the original configuration) leads to a significant improvement in detection performance, with a 3.2% increase in mAP. This comparison confirms that the IR modality carries more spatial information, while the RGB modality contains richer semantic information about the targets, thereby further validating the correctness of the module design.

TABLE VI
ABLATION STUDY ON THE FUSION STRATEGIES

NF	CWR	S2FM	Params	FLOPs	mAP (%)	mAP_{50} (%)
✓			35.1M	113.2G	65.0	80.7
✓	✓		**18.2M**	**52.2G**	64.5	80.3
✓		✓	38.1M	128.1G	68.3	83.1
✓	✓	✓	19.1M	57.7G	**69.0**	**83.3**

TABLE VII
ABLATION STUDY ON THE KERNEL SIZE USED IN THE SPATIAL EMBEDDING STEP

Kernel Size	Params	FLOPs	mAP (%)	mAP_{50} (%)
3×3	25.6M	94.5G	68.2	82.7
5×5	38.7M	167.2G	67.3	81.9
1×1	**18.2M**	**52.2G**	**69.0**	**83.3**

3) Ablation Study on the Fusion Strategies: This ablation study further investigates the contributions of different fusion configurations in the proposed SAFF framework to both model lightweight design and detection performance. We evaluate several fusion strategies, as summarized in Table VI. Specifically, "NF" denotes naive fusion, where duplicated IR and RGB extraction branches are employed without additional architectural enhancements. "CWR" refers to the channel width reduction strategy applied to the backbone for model lightweighting. "S2FM" indicates that feature fusion is performed using the proposed S2FM module; otherwise, basic feature concatenation is adopted.

As shown in Table VI, before applying the CWR strategy, naive fusion introduces substantial computational overhead, while simple modality concatenation provides only limited performance improvement. Moreover, even when the S2FM module is incorporated into the naive fusion setting, the detection performance remains inferior to that achieved by combining CWR with S2FM. This is mainly due to the redundancy in modality-specific feature extraction within the backbone when channel width reduction is not applied. Overall, these results further substantiate the rationality and effectiveness of the proposed fusion design.

4) Ablation Study on Spatial Embedding Kernel Size: We further conduct ablation experiments on the spatial embedding kernel size in the S2FM module by evaluating three representative configurations, i.e., 1×1, 3×3, and 5×5, corresponding to different receptive field scales. The experimental results are reported in Table VII. When the kernel size is set to 1×1, the proposed method achieves the best detection performance while also incurring the lowest computational cost in terms of both parameters and FLOPs.

This result is consistent with the design motivation of S2FM, where the spatial embedding step is intended to preserve shallow spatial positional information rather than perform deep spatial modeling. The 1×1 convolution preserves the original spatial structure of the feature maps without aggregating information from neighboring positions, allowing the module to maintain better positional cues. In contrast,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:02:06 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
COMPARISON OF THE COMMON FUSION STRATEGIES

Strategies	Params	FLOPs	mAP (%)	mAP_{50}(%)
Element-wise addition	18.3M	**52.2G**	64.1	80.4
Concat	**18.2M**	**52.2G**	64.5	80.3
Concat + SE	19.0M	52.8G	64.3	80.7
Concat + CBAM	18.6M	52.9G	65.1	80.5
Ours	19.1M	57.7G	**69.0**	**83.3**

TABLE IX
ABLATION STUDY ON THE NUMBER OF S2FMS

Number	Params	FLOPs	mAP (%)	mAP_{50} (%)
0	**18.2M**	**52.2G**	64.5	80.3
7	18.8M	57.0G	67.7	82.9
8	18.9M	57.5G	68.5	83.3
9	19.1M	57.7G	**69.0**	**83.3**
10	19.2M	58.0G	68.3	83.1

larger kernels (3×3, and 5×5) integrate information over a wider area, which may dilute these fine-grained spatial cues, slightly reducing detection accuracy. Thus, the ablation study indicates that a lightweight, shallow spatial embedding is more effective at preserving fine-grained positional cues, facilitating cross-modal feature interactions in S2FM.

5) Comparison of the Common Fusion Strategies: The effectiveness of the proposed S2FM for IR-RGB multimodal fusion is further evaluated through comparative experiments with several common fusion baselines. These baselines include simple element-wise addition and feature concatenation of the two modalities, as well as concatenation followed by lightweight attention modules such as SE [47] and CBAM [48].

For a fair comparison, the number of fusion operations and their insertion locations are kept identical across all baseline methods and S2FM. The experimental results are presented in Table VIII. The proposed S2FM consistently achieves superior detection performance compared with conventional fusion strategies, yielding an improvement of 3.9%–4.9% mAP. These findings demonstrate the effectiveness of S2FM in enabling robust and efficient IR-RGB feature fusion.

6) Ablation on the Number of S2FMs: Additionally, at the model architecture level, we conducted further ablation studies to examine the impact of the number of S2FM modules embedded within the network. The experimental results, as shown in Table IX, indicate that the detection performance improves as the number of S2FMs increases and reaches its optimum when nine S2FM modules are deployed across different scales. When more modules are added, the performance tends to plateau. This phenomenon may be attributed to overfitting and increased model complexity, which limit further performance improvements. Based on these observations, the final network architecture incorporates nine S2FMs.

7) Visualization and Interpretability Analysis: To further assess the interpretability and visual effectiveness of the proposed framework, we perform qualitative analyses through heatmap visualization and inspection of intermediate feature maps. These visual results provide intuitive insights into how SAFF jointly exploits spatial and semantic information from RGB and IR modalities, thereby facilitating a clearer understanding of its behavior in complex aerial scenes.

As illustrated in Fig. 7, under low-light conditions, the proposed method produces heatmaps with higher contrast than the single-modal baseline Yolo11m-obb as well as the multimodal methods C2Former and DMM. In dense vehicle scenarios, our model consistently generates more discriminative and compact feature responses. Specifically, in Fig. 7(a) and (b), SAFF demonstrates strong robustness to shadow-induced interference by concentrating its attention on densely distributed targets. In contrast, C2Former, DMM, and the single-modal baseline are more susceptible to low-light disturbances, leading to dispersed or misaligned regions of interest.

In Fig. 7(c), the proposed approach more effectively highlights individual objects within crowded regions, resulting in more accurate target localization and detection, which validates the effectiveness of the proposed strategy for aerial detection under challenging conditions. In Fig. 7(d), the comparative methods and the single-modal baseline are influenced by complex backgrounds, yielding overly diffuse attention maps and limited overlap with the true target regions. By comparison, the proposed SAFF model successfully suppresses background noise and maintains concentrated attention on relevant target areas, enabling precise detection even in highly cluttered scenes.

Finally, to further demonstrate the integration of target localization and semantic understanding, we visualized the model's feature maps and detection results in challenging scenarios. As depicted in Fig. 8, the proposed SAFF can effectively capture both the position and semantic attention of the objects to be detected, further improving detection performance.

V. DISCUSSION AND LIMITATION

In aerial object detection tasks, incorporating IR modality information has proven to be an effective strategy for enhancing detection performance under challenging conditions such as low illumination and smoke occlusion. However, existing methods often incur substantial computational overhead when integrating auxiliary modalities, which hinders their practical deployment. To address this issue, we proposed an SAFF that significantly reduces computational cost while maintaining strong detection accuracy, thereby achieving a better tradeoff between performance and efficiency.

Specifically, by analyzing the complementary characteristics of IR and visible modalities, we designed a semantic–positional complementary module that effectively extracts positional cues from the IR modality to enhance the semantically rich visible modality. Through multilayer cross-embedding, positional information is progressively integrated from shallow to deep layers, enabling improved fusion with lower computational complexity. The proposed method achieves SOTA performance across three widely used public IR-RGB detection datasets and yields consistent improvements on both Yolo11 and Yolo12 baselines, demonstrating its robustness and generalizability. We expect that this framework can serve as a strong baseline to advance multimodal IR-RGB detection research in the remote sensing community.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:02:06 UTC from IEEE Xplore. Restrictions apply.

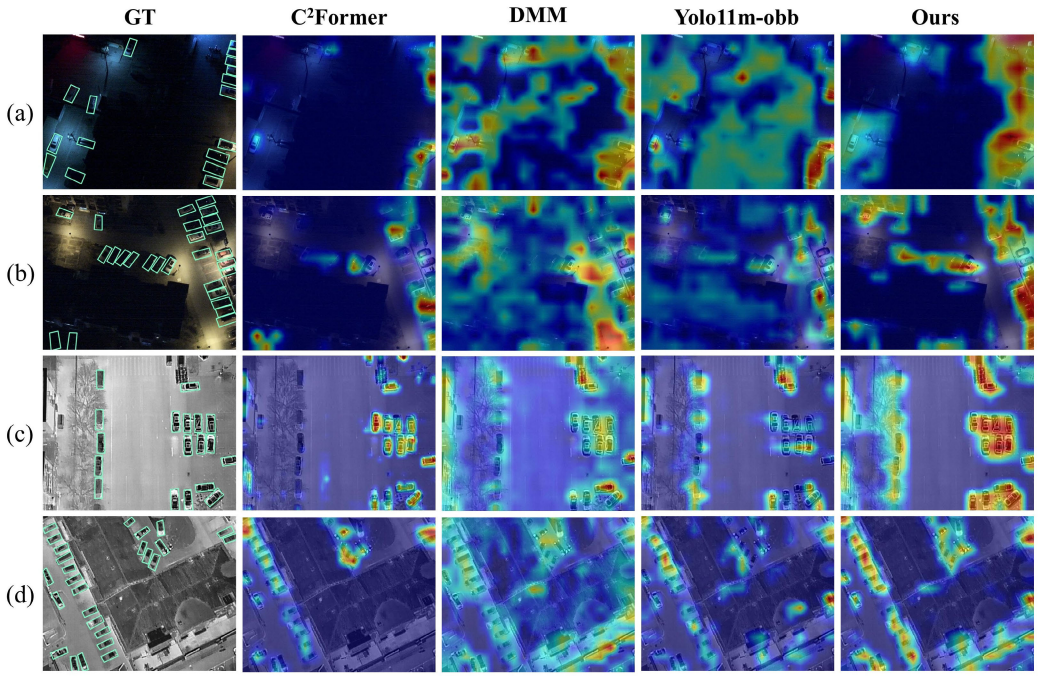


Fig. 7. Visualization of feature heatmaps for the compared methods and the proposed method.

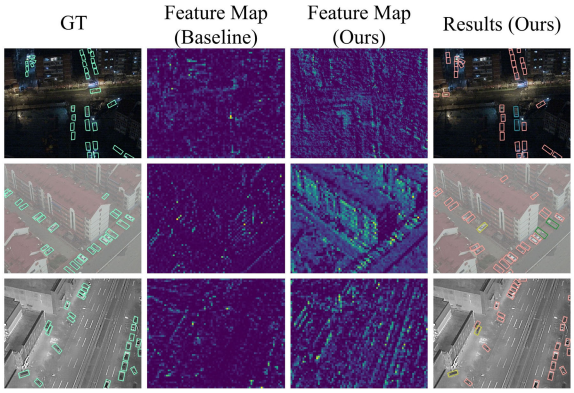


Fig. 8. Visualization of the feature map between the Baseline and the proposed model.

Despite its effectiveness, the framework has certain limitations. The current design is specifically optimized for IR-RGB data; extending it to other sensor pairs, such as LiDAR-RGB or SAR-RGB, may require minor architectural refinements. Moreover, while SAFF and S2FM substantially alleviate the computational burden, redundant processing may occur when the data volume from one modality is relatively small, leading to suboptimal resource utilization. Future work will focus on developing adaptive feature extraction strategies to dynamically balance computation across modalities and facilitate effective integration with various detector models.

VI. Conclusion

This article presents SAFF, an effective and efficient fusion-based detection framework, together with its core cross-modal fusion module, S2FM, to address the challenges of aerial multimodal detection. By effectively integrating semantic and spatial information from both RGB and IR modalities, SAFF achieves 2.6%–5.3% and 2.7%–7.1% improvements in mAP and mAP_{50} while reducing 12.2%–19.2% of computational cost compared with baseline models. Extensive experiments conducted on the DroneVehicle, VEDAI, and M3FD datasets, using Yolo11 and Yolo12 as baselines, demonstrate that SAFF consistently outperforms state-of-the-art methods, achieving an optimal balance between accuracy and efficiency. We anticipate that SAFF can serve as a strong benchmark framework for multimodal aerial detection tasks and foster further research on IR-RGB fusion-based detection methods in the field of remote sensing and intelligent surveillance.

References

[1] Y. Zhang, W. Guo, C. Wu, W. Li, and R. Tao, "FANet: An arbitrary direction remote sensing object detection network based on feature fusion and angle classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5608811, doi: 10.1109/TGRS.2023.3273354.

[2] Y. Shen, D. Liu, F. Zhang, and Q. Zhang, "Fast and accurate multi-class geospatial object detection with large-size remote sensing imagery using CNN and truncated NMS," *ISPRS J. Photogramm. Remote Sens.*, vol. 191, pp. 235–249, Sep. 2022.

[3] X. Xie, G. Cheng, C. Rao, C. Lang, and J. Han, "Oriented object detection via contextual dependence mining and penalty-incentive allocation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5618010, doi: 10.1109/TGRS.2024.3385985.

[4] R. Zhang et al., "A benchmark and frequency compression method for infrared few-shot object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5001711, doi: 10.1109/TGRS.2025.3540945.

[5] R. Zhang, L. Xu, Z. Yu, Y. Shi, C. Mu, and M. Xu, "Deep-IRTarget: An automatic target detector in infrared imagery using dual-domain feature extraction and allocation," *IEEE Trans. Multimedia*, vol. 24, pp. 1735–1749, 2022.

[6] S. Dong et al., "SeaDATE: Remedy dual-attention transformer with semantic alignment via contrast learning for multimodal object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 5, pp. 4713–4726, May 2025.

[7] X. Xie et al., "Learning discriminative representation for fine-grained object detection in remote sensing images," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 8, pp. 8197–8208, Aug. 2025.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:02:06 UTC from IEEE Xplore. Restrictions apply.

[8] R. Zhang, Z. Cao, Y. Huang, S. Yang, L. Xu, and M. Xu, "Visible-infrared person re-identification with real-world label noise," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 5, pp. 4857–4869, May 2025.

[9] W. Bao, M. Huang, J. Hu, and X. Xiang, "Dual-dynamic cross-modal interaction network for multimodal remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5401013, doi: 10.1109/TGRS.2025.3530085.

[10] X. Yuan, G. Cheng, R. Yao, and J. Han, "Semantic differentiation aids oriented small object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 6, pp. 5966–5979, Jun. 2025.

[11] Y. Chen, Z. Ye, H. Sun, T. Gong, S. Xiong, and X. Lu, "Global–local fusion with semantic information guidance for accurate small object detection in UAV aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4701115, doi: 10.1109/TGRS.2025.3532612.

[12] B. Yang, H. Xiao, and B. Zhang, "A unified object and keypoint detection framework for personal protective equipment use," *Develop. Built Environ.*, vol. 20, Dec. 2024, Art. no. 100559.

[13] S. Zhao, H. Chen, X. Zhang, P. Xiao, L. Bai, and W. Ouyang, "RS-Mamba for large remote sensing image dense prediction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5633314, doi: 10.1109/TGRS.2024.3425540.

[14] Y. Xiao, T. Xu, Y. Xin, and J. Li, "FBRT-YOLO: Faster and better for real-time aerial image detection," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2025, vol. 39, no. 8, pp. 8673–8681.

[15] X. Xie, C. Lang, S. Miao, G. Cheng, K. Li, and J. Han, "Mutual-assistance learning for object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 12, pp. 15171–15184, Dec. 2023.

[16] X. Yang, J. Yan, W. Liao, X. Yang, J. Tang, and T. He, "SCRDet++: Detecting small, cluttered and rotated objects via instance-level feature denoising and rotation loss smoothing," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 2384–2399, Feb. 2023.

[17] H. Xiao, B. Yang, Y. Lu, W. Chen, S. Lai, and B. Gao, "Automated detection of complex construction scenes using a lightweight transformer-based method," *Autom. Construction*, vol. 177, Sep. 2025, Art. no. 106330.

[18] Y. Sun, B. Cao, P. Zhu, and Q. Hu, "Drone-based RGB-infrared cross-modality vehicle detection via uncertainty-aware learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 10, pp. 6700–6713, Oct. 2022.

[19] F. Lin, C. Gao, and K. D. Yamada, "An effective convolutional neural network for visualized understanding transboundary air pollution based on himawari-8 satellite images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[20] H. Xiao, B. Yang, Y. Yu, J. Han, Z. Lian, and S. Lai, "Orientation-aware detection system for real-time monitoring of cracks in steel structures," *Expert Syst. Appl.*, vol. 299, Mar. 2026, Art. no. 129932. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S095741742503547X

[21] M. Yuan and X. Wei, "C^2former: Calibrated and complementary transformer for RGB-infrared object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5403712, doi: 10.1109/TGRS.2024.3376819.

[22] R. Zhang et al., "Differential feature awareness network within antagonistic learning for infrared-visible object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 8, pp. 6735–6748, Aug. 2024.

[23] G. Lv, X. Gao, A. Dong, Z. Wei, and J. Cheng, "SLFusion: A structure-aware infrared and visible image fusion network for low-light scenes," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 12, pp. 11877–11892, Dec. 2025, doi: 10.1109/TCSVT.2025.3586679.

[24] Y. Zhao, Q. Zheng, P. Zhu, X. Zhang, and W. Ma, "TUFusion: A transformer-based universal fusion algorithm for multimodal images," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 3, pp. 1712–1725, Mar. 2024.

[25] M. Zhou et al., "DMM: Disparity-guided multispectral mamba for oriented object detection in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2024, Art. no. 5404913, doi: 10.1109/TGRS.2025.3578309.

[26] S. Li, Y. Li, S. Niu, J. Yang, X. Yang, and X. Yue, "BDSFusion: A bidirectionally driven saliency fusion network for enhancing small target detection in infrared dual-band images," *Inf. Fusion*, vol. 126, Feb. 2026, Art. no. 103600.

[27] K. He, X. Zhang, S. Ren, and J. Sun, "Spatial pyramid pooling in deep convolutional networks for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 9, pp. 1904–1916, Sep. 2015.

[28] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3500–3509.

[29] X. Xie, G. Cheng, Q. Li, S. Miao, K. Li, and J. Han, "Fewer is more: Efficient object detection in large aerial images," *Sci. China Inf. Sci.*, vol. 67, no. 1, Jan. 2024, Art. no. 112106, doi: 10.1007/s11432-022-3718-5.

[30] J. Zhao, Z. Ding, Y. Zhou, H. Zhu, W.-L. Du, and R. Yao, "ReDiffDet: Rotation-equivariant diffusion model for oriented object detection," in *Proc. Comput. Vis. Pattern Recognit. Conf.*, Jun. 2025, pp. 24429–24439.

[31] H. Xu et al., "Rethinking boundary discontinuity problem for oriented object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 17406–17415.

[32] J. Zhao et al., "OrientedFormer: An end-to-end transformer-based oriented object detector in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5640816, doi: 10.1109/TGRS.2024.3456240.

[33] J. Murrugarra-LLerena, J. Henrique Lima Marques, and C. R. Jung, "GauCho: Gaussian distributions with Cholesky decomposition for oriented object detection," 2025, *arXiv:2502.01565*.

[34] G. Jocher, J. Qiu, and A. Chaurasia. (Jan. 2023). *Ultralytics YOLO*. [Online]. Available: https://github.com/ultralytics/ultralytics

[35] Y. Tian, Q. Ye, and D. Doermann, "YOLOv12: Attention-centric real-time object detectors," 2025, *arXiv:2502.12524*.

[36] Y. Fu, W. Li, S. Fan, Y. Jiang, and H. Bai, "CAL-Net: Conditional attention lightweight network for in-orbit landslide detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4408515.

[37] S. Razakarivony and F. Jurie, "Vehicle detection in aerial imagery: A small target detection benchmark," *J. Vis. Commun. Image Represent.*, vol. 34, pp. 187–203, Jan. 2016.

[38] J. Liu et al., "Target-aware dual adversarial learning and a multi-scenario multi-modality benchmark to fuse infrared and visible for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 5792–5801.

[39] Y. Zhao et al., "DETRs beat YOLOs on real-time object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 16965–16974.

[40] W. Lv, Y. Zhao, Q. Chang, K. Huang, G. Wang, and Y. Liu, "RT-DETRv2: Improved baseline with bag-of-freebies for real-time detection transformer," 2024, *arXiv:2407.17140*.

[41] S. Huang, Z. Lu, X. Cun, Y. Yu, X. Zhou, and X. Shen, "DEIM: DETR with improved matching for fast convergence," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 15162–15171.

[42] Y. Peng, H. Li, P. Wu, Y. Zhang, X. Sun, and F. Wu, "D-FINE: Redefine regression task in DETRs as fine-grained distribution refinement," 2024, *arXiv:2410.13842*.

[43] Y. Xiao, F. Meng, Q. Wu, L. Xu, M. He, and H. Li, "GM-DETR: Generalized muiltispectral detection transformer with efficient fusion encoder for visible-infrared detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2024, pp. 5541–5549.

[44] J. Shen, Y. Chen, Y. Liu, X. Zuo, H. Fan, and W. Yang, "ICAFusion: Iterative cross-attention guided feature fusion for multispectral object detection," *Pattern Recognit.*, vol. 145, Jan. 2024, Art. no. 109913.

[45] K. Hu, Y. He, Y. Li, J. Zhao, S. Chen, and Y. Kang, "EI^{2}det: Edge-guided illumination-aware interactive learning for visible-infrared object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 7, pp. 7101–7115, Jul. 2025.

[46] M. Yuan, X. Shi, N. Wang, Y. Wang, and X. Wei, "Improving RGB-infrared object detection with cascade alignment-guided transformer," *Inf. Fusion*, vol. 105, May 2024, Art. no. 102246.

[47] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[48] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. ECCV*, 2018, pp. 3–19.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:02:06 UTC from IEEE Xplore. Restrictions apply.